

Physical Lithium-Ion Battery Model

Mathematical Formulation and MATLAB/Simulink Implementation

FYP Battery Digital Twin Project

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Abstract

This document describes the physical lithium-ion battery plant developed for the Battery Digital Twin project. The model combines state-of-charge (SOC), open-circuit voltage (OCV), two polarization RC branches, lumped thermal dynamics, capacity degradation, internal-resistance growth, and terminal-voltage calculation. The physical plant is kept separate from the BMS, sensors, state estimation, AI/ML, optimization, and control layers so that the same architecture can later be parameterized with experimentally measured battery data.

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1 Purpose and Role of the Physical Model

The physical battery is the plant at the bottom of the digital-twin architecture:

Current Command $\rightarrow$ Physical Battery $\rightarrow$ Sensors $\rightarrow$ BMS $\rightarrow$ State Estimation $\rightarrow$ Digital Twin/AI/O (1)

The model is an equivalent-circuit battery model rather than a full electrochemical model. This was selected because it provides physically interpretable states with relatively low computational cost and is suitable for real-time BMS and digital-twin development.

2 Inputs and Outputs

The top-level `Physical_Battery` subsystem has two inputs:

Input	Symbol	Meaning
Current command	I_{command}	Applied battery current request. Positive is discharge; negative is charge.
Ambient temperature	T_a	Temperature surrounding the battery.

Table 1: Physical battery inputs.

The main outputs are V_{battery} , SOC_{true} , T_b , SOH_{true} , $R_{0,\text{current}}$, $Q_{\text{available}}$, $Q_{\text{Ah,throughput}}$, and I_{battery} .

The sign convention is

$$I > 0 \Rightarrow \text{discharge}, \quad I < 0 \Rightarrow \text{charge}. \quad (2)$$

3 Parameterization

The battery-dependent parameters are loaded from the selected battery profile. The current project includes a generic profile and a Molicel INR-18650-M35A profile. The current Molicel cell specifications are:

Parameter	Value
Nominal capacity Q_{nom}	3.45 Ah
Nominal voltage	3.60 V
Maximum voltage	4.20 V
Minimum voltage	2.50 V
Maximum charge current	1.70 A
Maximum discharge current	10.00 A
Initial SOC	0.95

Table 2: Current Molicel profile specifications.

The ECM, thermal, OCV and aging parameters are currently model parameters and should be identified/refined experimentally before making manufacturer-specific performance claims.

4 Equivalent-Circuit Model

The electrical model consists of an SOC-dependent OCV source, a series ohmic resistance R_0 , and two first-order polarization branches (R_1, C_1) and (R_2, C_2) . The terminal voltage is

$$V_{\text{terminal}} = V_{\text{OCV}}(\text{SOC}) - IR_{0,\text{current}} - V_1 - V_2. \quad (3)$$

5 Hand-Drawn Circuit Diagram

Insert hand-drawn equivalent-circuit diagram here

Draw or insert the equivalent circuit showing: (1) $V_{\text{OCV}}(\text{SOC})$, (2) R_0 , (3) the R_1 – C_1 branch, (4) the R_2 – C_2 branch, (5) current direction, and (6) terminal-voltage polarity. A scanned/photo image can replace this box using `\includegraphics`.

Suggested caption: *Equivalent-circuit representation of the lithium-ion battery used in the physical plant model.*

6 SOC Model

SOC is normalized remaining charge:

$$0 \leq \text{SOC} \leq 1. \quad (4)$$

The model uses Coulomb counting:

$$\frac{d\text{SOC}}{dt} = -\frac{\eta_I I}{3600 Q_{\text{available}}}. \quad (5)$$

Starting from charge balance $dQ/dt = -\eta_I I/3600$ and $\text{SOC} = Q/Q_{\text{available}}$ gives Equation (5). The 3600 converts amperes to Ah/s. The SOC state is saturated to $[0, 1]$ in Simulink.

7 OCV–SOC Model

The open-circuit voltage is represented by a nonlinear lookup relationship:

$$V_{\text{OCV}} = f(\text{SOC}). \quad (6)$$

A one-dimensional lookup table maps SOC breakpoints to OCV values. This approach was selected because lithium-ion OCV is nonlinear with SOC and experimental OCV–SOC data can later replace the current model table without changing the architecture.

8 First RC Polarization Model

The first dynamic branch is

$$\boxed{\frac{dV_1}{dt} = \frac{I}{C_1} - \frac{V_1}{R_1 C_1}} \quad (7)$$

Equivalently, $R_1 C_1 (dV_1/dt) + V_1 = R_1 I$. Its time constant is $\tau_1 = R_1 C_1$. Current model values are $R_1 = 0.020 \Omega$ and $C_1 = 2200 \text{ F}$, giving $\tau_1 = 44 \text{ s}$. Initial condition: $V_1(0) = 0$.

9 Second RC Polarization Model

The second dynamic branch is

$$\boxed{\frac{dV_2}{dt} = \frac{I}{C_2} - \frac{V_2}{R_2 C_2}} \quad (8)$$

Its time constant is $\tau_2 = R_2 C_2$. Current model values are $R_2 = 0.012 \Omega$ and $C_2 = 15\,000 \text{ F}$, giving $\tau_2 = 180 \text{ s}$. Initial condition: $V_2(0) = 0$.

10 Terminal Voltage

The terminal voltage combines equilibrium voltage, instantaneous ohmic drop, and the two polarization voltages:

$$\boxed{V_{\text{terminal}} = V_{\text{OCV}}(\text{SOC}) - IR_{0,\text{current}} - V_1 - V_2} \quad (9)$$

Thus, the model captures both instantaneous and dynamic voltage response.

11 Thermal Model

The lumped thermal energy balance is

$$\boxed{C_{th} \frac{dT_b}{dt} = P_{gen} - P_{loss}} \quad (10)$$

The implemented heat-generation model is

$$\boxed{P_{gen} = I^2 R_{0,\text{current}} + \frac{V_1^2}{R_1} + \frac{V_2^2}{R_2}} \quad (11)$$

Heat loss to ambient is

$$P_{loss} = hA(T_b - T_a). \quad (12)$$

Therefore,

$$\frac{dT_b}{dt} = \frac{I^2 R_{0,current} + V_1^2/R_1 + V_2^2/R_2 - hA(T_b - T_a)}{C_{th}}. \quad (13)$$

The initial condition is $T_b(0) = T_{initial}$. The physical temperature state is not saturated; temperature limits are handled by the BMS protection layer.

12 Aging and SOH Model

The aging model represents capacity fade and internal-resistance growth. Both charging and discharging contribute to cycling stress, so absolute current is used.

12.1 Ah Throughput

$$\frac{dQ_{Ah,throughput}}{dt} = \frac{|I|}{3600}. \quad (14)$$

Hence,

$$Q_{Ah,throughput}(t) = \int_0^t \frac{|I(\tau)|}{3600} d\tau. \quad (15)$$

12.2 Temperature and SOC Stress

The current implementation uses lookup-table stress factors

$$F_T = f_T(T_b), \quad F_{SOC} = f_{SOC}(SOC), \quad (16)$$

and combines them as

$$F_{stress} = F_T F_{SOC}. \quad (17)$$

Current placeholder temperature breakpoints are $[0, 10, 20, 25, 30, 35, 40, 45]^\circ C$ with stress factors $[1.5, 1.2, 1.05, 1.0, 1.05, 1.2, 1.5, 2.0]$. Current SOC breakpoints are $[0, 0.1, 0.2, 0.4, 0.6, 0.8, 0.9, 1.0]$ with factors $[1.5, 1.2, 1.05, 1.0, 1.0, 1.05, 1.2, 1.5]$. These are model-development assumptions, not universal experimental laws.

12.3 Capacity Fade and SOH

With capacity-fade coefficient k_Q ,

$$\frac{dD_Q}{dt} = k_Q F_{stress} \frac{|I|}{3600}. \quad (18)$$

The physical SOH is

$$SOH = 1 - D_Q, \quad 0 \leq SOH \leq 1. \quad (19)$$

Available capacity is therefore

$$Q_{available} = SOH Q_{nom}. \quad (20)$$

This creates a feedback from aging into the SOC dynamics.

12.4 Internal Resistance Growth

With resistance-growth coefficient k_R ,

$$\frac{dD_R}{dt} = k_R F_{stress} \frac{|I|}{3600}. \quad (21)$$

The dynamic resistance is

$$R_{0,current} = R_0(1 + D_R). \quad (22)$$

The resulting resistance feeds both terminal-voltage and thermal calculations.

13 Coupling of the Submodels

The principal feedback relationships are

$$I \rightarrow SOC \rightarrow V_{OCV}, \quad (23)$$

$$I \rightarrow V_1, V_2 \rightarrow V_{terminal}, \quad (24)$$

$$I, V_1, V_2, R_0 \rightarrow P_{gen} \rightarrow T_b, \quad (25)$$

$$T_b, SOC, I \rightarrow F_{stress} \rightarrow SOH, \quad (26)$$

$$SOH \rightarrow Q_{available} \rightarrow SOC \text{ dynamics}. \quad (27)$$

Thus the plant is a coupled electrical–thermal–aging system rather than independent submodels.

14 Simulink Submodel Structure

The `Physical_Battery` subsystem contains:

1. SOC Model;
2. OCV Model;
3. R1C1 Model;
4. R2C2 Model;
5. Thermal Model;
6. Aging Model;

7. Terminal Voltage Model.

The principal signal flow is:

```

I_command -> SOC Model
           -> R1C1 Model
           -> R2C2 Model
           -> Thermal Model
           -> Aging Model
           -> Terminal Voltage Model

SOC_true -> OCV Model -> Terminal Voltage
Q_available -> SOC Model
T_ambient -> Thermal Model
T_battery -> Aging Model
R0_current -> Terminal Voltage and Thermal Model
V1, V2 -> Terminal Voltage and Thermal Model

```

15 Initial Conditions

The physical plant uses

$$SOC(0) = SOC_{initial}, \quad (28)$$

$$V_1(0) = 0, \quad V_2(0) = 0, \quad (29)$$

$$T_b(0) = T_{initial}, \quad (30)$$

$$SOH(0) = 1, \quad (31)$$

$$D_Q(0) = 0, \quad D_R(0) = 0. \quad (32)$$

Consequently,

$$Q_{available}(0) = Q_{nom}, \quad R_{0,current}(0) = R_0. \quad (33)$$

16 Why Each Submodel Is Included

Submodel	Purpose
SOC	Tracks remaining charge and provides the principal battery state.
OCV	Converts SOC into equilibrium voltage.
R_1C_1	Represents faster transient/polarization voltage dynamics.
R_2C_2	Represents slower transient/polarization voltage dynamics.
Thermal	Represents temperature rise and heat exchange with ambient.
Aging	Represents capacity fade and resistance growth with use and stress.
Terminal voltage	Produces the measurable battery voltage from OCV, ohmic and polarization terms.

Table 3: Purpose of the physical submodels.

17 Relationship to Sensors and BMS

The physical plant generates true quantities $V_{battery}$, $I_{battery}$, T_b , SOC_{true} and SOH_{true} . The sensor layer then adds measurement characteristics such as bias and noise. The BMS operates on measured signals rather than directly accessing the true internal states. This separation allows validation of estimation error and later replacement of simulated measurements by experimental sensor data.

18 Assumptions and Limitations

1. The plant represents a lumped single cell.
2. OCV is represented as a function of SOC.
3. Electrical dynamics use two first-order RC branches.
4. Temperature is represented by one lumped thermal state.
5. Aging is represented by empirical stress factors based on temperature and SOC.
6. Capacity fade and resistance growth are cumulative model states.
7. ECM, thermal, OCV and aging parameters are currently model parameters and require experimental identification/refinement.

Therefore the model should be described as a **physics-informed equivalent-circuit battery model**, not as a complete electrochemical model.

19 Future Parameter Identification

When real battery data become available, the following should be identified/refined: OCV–SOC curve; R_0, R_1, C_1, R_2, C_2 ; SOC/temperature dependence of ECM parameters; C_{th} and hA ; k_Q and k_R ; and experimentally supported temperature/SOC aging relationships. The intended workflow is

Battery Cycling → Data Acquisition → Parameter Identification → Model Update → Validation. (34)

20 Complete Governing Equations

For reference, the current plant can be summarized by

$$\frac{dSOC}{dt} = -\frac{\eta_I I}{3600 Q_{available}}, \quad (35)$$

$$V_{OCV} = f(SOC), \quad (36)$$

$$\frac{dV_1}{dt} = \frac{I}{C_1} - \frac{V_1}{R_1 C_1}, \quad (37)$$

$$\frac{dV_2}{dt} = \frac{I}{C_2} - \frac{V_2}{R_2 C_2}, \quad (38)$$

$$P_{gen} = I^2 R_{0,current} + V_1^2 / R_1 + V_2^2 / R_2, \quad (39)$$

$$P_{loss} = hA(T_b - T_a), \quad (40)$$

$$\frac{dT_b}{dt} = \frac{P_{gen} - P_{loss}}{C_{th}}, \quad (41)$$

$$\frac{dQ_{Ah,throughput}}{dt} = \frac{|I|}{3600}, \quad (42)$$

$$F_{stress} = F_T F_{SOC}, \quad (43)$$

$$\frac{dD_Q}{dt} = k_Q F_{stress} \frac{|I|}{3600}, \quad (44)$$

$$SOH = 1 - D_Q, \quad (45)$$

$$Q_{available} = SOH Q_{nom}, \quad (46)$$

$$\frac{dD_R}{dt} = k_R F_{stress} \frac{|I|}{3600}, \quad (47)$$

$$R_{0,current} = R_0(1 + D_R), \quad (48)$$

$$V_{terminal} = V_{OCV} - I R_{0,current} - V_1 - V_2. \quad (49)$$

21 Conclusion

The physical battery model provides the plant for the FYP Battery Digital Twin. It combines SOC, OCV, dynamic electrical polarization, temperature, capacity degradation, and resistance growth in a modular structure. The architecture is intentionally parameterized so experimentally identified battery data can later replace the current model parameters without redesigning the overall Simulink structure.